Correlation of alloy microstructure with oxidation behavior in chromia-forming intermetallic-reinforced Cr alloys

M.P. Brady, P.F. Tortorelli, L.R. Walker
Oak Ridge National Laboratory, Oak Ridge, TN 37831-6156, USA
E-mail: bradymp@ornl.gov

The oxidation behavior of the Cr(X) solid solution and intermetallic Cr_2X Laves phases (X = Nb, Ta) was studied individually and in combination at 1,100°C in humid air. The two-phase Cr(Ta) + Cr_2Ta alloys exhibited superior oxidation resistance to either the single-phase Cr(Ta) or the single-phase Cr_2Ta alloys. However, the two-phase Cr(Nb) + Cr_2Nb alloys exhibited lower oxidation resistance than either the single-phase Cr(Nb) or Cr_2Nb alloys. The two-phase Cr(Ta) + Cr_2Ta alloys were also found to exhibit superior oxidation resistance to the two-phase Cr(Nb) + Cr_2Nb alloys. This was hypothesized to be related to the lower amount of the Cr_2X phase in the Cr(Ta)–Cr_2Ta eutectic and to the lower solubility of Ta in the Cr(X) phase, relative to the Cr–Nb system. Both of these factors resulted in significantly less oxidation and nitridation in the Cr(Ta) + Cr_2Ta alloys.

Keywords: Cr, Laves phase, chromia, nitride, oxidation

INTRODUCTION

Laves phase strengthened Cr alloys are being developed for use at high-temperatures [1–9]. Such alloys have high melting points, excellent high-temperature strength, and potential for high-temperature oxidation and corrosion resistance via the formation of a protective Cr_2O_3-based scale. The Cr(X)–Cr_2X systems of interest (X = Hf, Nb, Ta, Zr, …) typically contain eutectic reactions. Accordingly, the microstructure of these alloys consist of Cr_2X Laves phase particles or lamellae dispersed in a Cr(X) solid solution matrix. The Cr_2X Laves phase particles provide high-temperature strength, and the Cr(X) phase is a potential source of ductility and oxidation/corrosion resistance. Binary alloys based on the Cr(Ta)–Cr_2Ta eutectic were recently shown to exhibit superior oxidation resistance to similar Cr(Nb)–Cr_2Nb alloys in a six cycle, 1,100°C screening test in humid air (Figure 1). Given the similarities between Nb and Ta, the superior oxidation resistance of the Cr(Ta)-Cr_2Ta alloys is surprising. To gain an understanding of this effect, the oxidation behavior of the Cr(X) solid solution phase and Cr_2X Laves phase (X = Nb, Ta) are under investigation individually and in combination from 950–1,100°C in dry and humid air [11]. The present work reports on correlation of alloy microstructure with oxidation behavior at 1,100°C in humid air.

EXPERIMENTAL

Small 50–60 g castings of Cr(X), Cr_2X, and Cr(X) + Cr_2X alloys were prepared by arc melting and drop casting in a chilled copper mold. The nominal compositions (atomic percent, at.%) of the alloys chosen for study were:

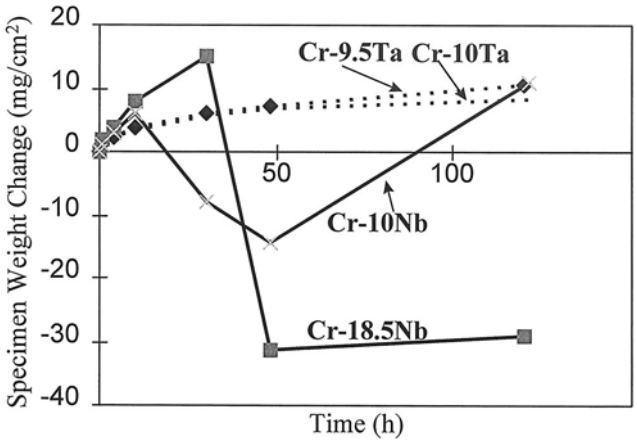


Figure 1 Cyclic oxidation at 1,100°C in humid "room" air [9,10]. At each data point the sample was air cooled, weighed, and returned to the furnace at temperature. The Cr–10Nb and Cr–10Ta alloys were tested as-cast and the Cr–18.5Nb and Cr–9.5Ta alloys were heat treated at 1300°C for 24 h prior to testing (at.%).

0964-3409/00/01/00035–12
© 2000 Science Reviews

Cr–2.5Nb and Cr–1Ta to represent the Cr(X) phase (Nb has a higher solubility in Cr than Ta), Cr–1Nb for comparative purposes with Cr–1Ta, Cr–31Nb and Cr–31Ta to represent the Cr_2X Laves phase, and the respective two-phase Cr(X)–Cr_2X eutectic compositions Cr–18.5Nb and Cr–9.5Ta (Figure 2). The eutectic compositions were of particular interest for understanding the oxidation behavior of two-phase Cr(X) + Cr_2X alloys because the development effort is focused on hypoeutectic alloys for mechanical property reasons, and the microstructures of interest therefore typically consist of primary Cr(X) phase and regions of Cr(X)–Cr_2X eutectic. Unless otherwise noted, the alloy castings were heat treated for 24 h at 1,300°C in vacuum.

Disk shaped specimens of approximately 12–13 mm in diameter and 1–1.5 mm thickness were polished to a 600 grit finish using SiC paper. Oxidation behavior was studied by thermogravimetric analysis (TGA) in 24 h tests at 1,100°C open to "room" air of varying daily humidity (typically 50–80%). Select samples were also placed in alumina crucibles (with lids) and isothermally or cyclically exposed (air cooled to room-temperature after 1, 4, 10, 30, 48, 120 cumulative hours) at 1,100°C in a tube furnace open to room air for periods up to a week. As expected for this temperature, there was some CrO_3 volatility, which was evident by green "stains" on the inside of the lids. However, volatility-related weight loss did not appear to be significant compared to oxidative weight gains under the conditions studied. Oxidized coupons were characterized by X-ray diffraction (XRD), light microscopy, secondary mode scanning electron microscopy (SEM), and electron probe microanalysis (EPMA) using pure element standards for Cr, Nb, Ta; a BN standard for N; and an Al_2O_3 standard for O. All EPMA measurements are reported in at.%.

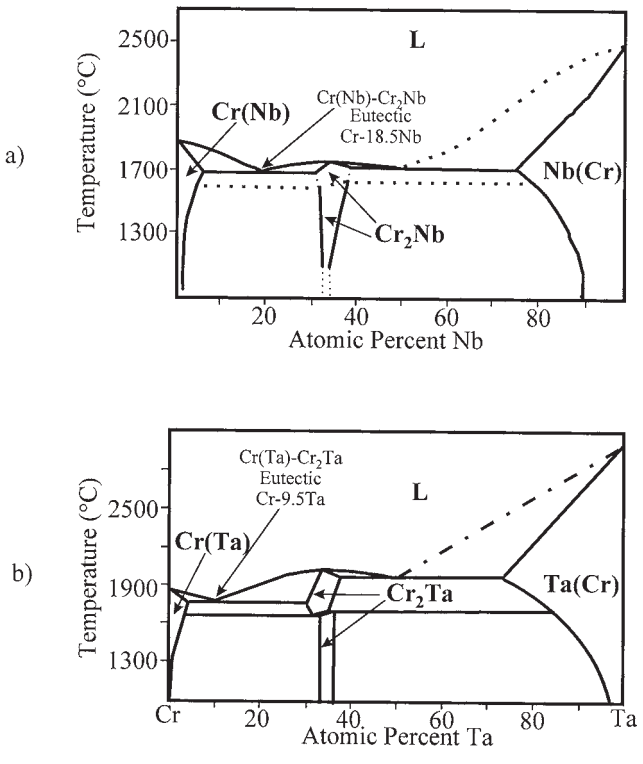


Figure 2 Schematic phase diagrams used for alloy selection. (a) Cr–Nb (based on reference 12), (b) Cr–Ta (based on [13,14]).

The commercial purity Cr source used to make the alloys was found to contain high levels of oxygen (≈ 7,000 weight parts per million, wppm) and sulfur (≈ 350 wppm). The high impurity levels had little effect on the Cr-Ta alloys under the conditions studied. However, a beneficial effect of a high level of oxygen and sulfur impurities was observed for the Cr_2Nb-based alloys but only in filtered compressed dry air. These results are discussed elsewhere [11].

RESULTS

Single-phase Cr(X)

Initially, Cr–1Nb and Cr–1Ta were chosen to represent the Cr solid solution phase. However, the respective phase diagrams (Figure 2) suggest that Nb has a higher solubility in Cr than Ta. Experimental determination of the solubility limit for a given temperature is difficult due to the decreasing solubility of Nb or Ta with decreasing temperature below the eutectic and the precipitation of the Laves phase. EPMA analysis of air-cooled samples after exposure from 1,100–1,300°C suggested that Cr exhibited a Nb solubility of approximately 1.5 at.% and a Ta solubility of approximately 0.6 at.%. The compositions of the model Cr solid solution alloys were selected to fall just past the solubility limit to ensure that component activities would be the same as in the two-phase alloys. Therefore, Cr–2.5Nb was selected to represent the Cr(Nb) phase and Cr–1Ta to represent the Cr(Ta) phase. The Cr–1Nb alloy was studied for comparative purposes with Cr–1Ta on a strictly concentration basis.

The Cr–1Nb and Cr–1Ta alloys exhibited similar, nearly parabolic oxidation kinetics, although a discontinuity typical of scale cracking was also observed for Cr–1Ta (Figure 3). The rate for Cr–2.5Nb was faster than those of Cr–1Nb and Cr–1Ta. The TGA curve for one Cr–2.5Nb sample showed a discontinuity typical of scale cracking followed by a rapid rate of weight gain- much more rapid than after the discontinuity observed in the TGA data for Cr–1Ta.

Extensive scale spallation during air cooling to room-temperature was observed for all three model Cr solid solution alloys. Chromia scales approximately 20–25 μm thick formed on Cr–1Nb and Cr–1Ta after 24 h at 1,100°C in humid air (Figure 4 a,b). The surface of the scale on Cr–1Nb was locally convoluted and, thus, highly friable (Figure 4a). Such local friability led to wide bands of disconnected Cr_2O_3 particles on top of the scale such as is evident in Figure 4a. Similar regions were observed on most alloys studied, although it appeared to be more prevalent for Cr–Nb than Cr–Ta alloys.

A Cr_2O_3 scale was also formed on Cr–2.5Nb. However, variation in scale features was observed within a given sample and from sample to sample. Scale regions similar to those on Cr–1Nb were observed on Cr–2.5Nb, although both the friable and dense Cr_2O_3 regions were somewhat thicker after the 24 h/1,100°C exposure (Figure 4c- lower weight gain Cr–2.5Nb TGA test sample from Figure 3). However, an as-cast sample of Cr–2.5Nb exposed for 24 h at 1,100°C showed extensive subscale nitridation of Cr to a depth of 40–50 μm (Figure 4 d). The composition of the nitride was approximately 61Cr–1Nb-38N. (The weight gain for this sample

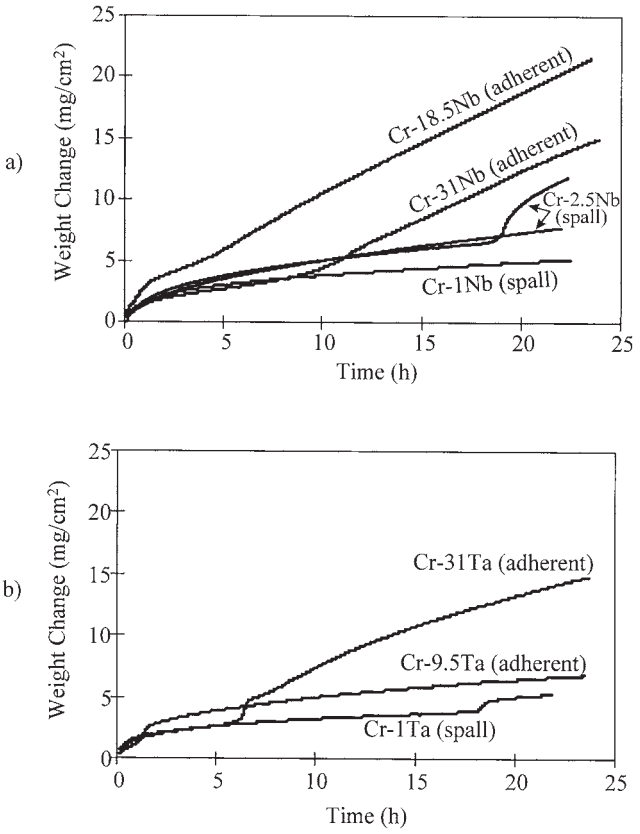


Figure 3 TGA data at 1,100°C in humid air (varying daily humidity of 50–80%). (a) Cr–Nb alloys (b) Cr–Ta alloys.

fell between the two TGA tests for Cr–2.5Nb shown in Figure 3). The Cr_2O_3 scale formed on as-cast Cr-2.5Nb was porous. EDS spot analysis of oxide immediately adjacent to the porous regions showed high levels of Nb, presumably indicating the presence of Nb-rich oxide. Similar regions of extensive nitridation as well as local regions of scale porosity were also observed in the Cr–2.5Nb sample shown in Figure 4c, primarily in and near edge regions of the coupon. Little nitridation was observed for Cr–1Nb or Cr–1Ta after 24 h at 1,100°C in humid air, although it should be noted that longer term exposures at this temperature leads to extensive nitridation in these alloys. The nitride composition was similar to that measured in Cr–2.5Nb

Single-phase Cr_2X Laves phase

Both Cr–31Nb and Cr–31Ta showed a transition to rapid oxidation kinetics after 5–10 h at 1,100°C in humid air (Figure 3). The scale formed on Cr-31Nb consisted of a layer of Cr_2O_3 approximately 15–20 μm thick (including an outer friable layer of disconnected Cr_2O_3 particles likely formed during metallographic preparation) overlying a region of mixed Cr + Nb oxides approximately 50–60 μm thick (Figure 5a) . Nitridation on the order of 100–150 μm deep was observed underneath the scale, and generally fell within the composition range of (40–50)Cr–(20–25)Nb-(25–35)N. The scale formed on Cr–31Ta consisted of alternating layers of Cr_2O_3 and Cr + Ta rich oxide (Figure 5b). Nitridation below the scale on the order of 100–150 μm deep was also evident. The composition of the nitrided Cr_2Ta Laves phase was approximately 40Cr–20Ta–40N.

Two-phase Cr(X) + Cr_2X eutectic alloys

The Cr(Nb)–Cr_2Nb eutectic alloy, Cr–18.5Nb, exhibited rapid linear oxidation kinetics similar to that for Cr–31Nb (Figure 3). However, unlike Cr–31Nb, Cr–18.5Nb did not show an initial parabolic region; thus the overall weight gain after 24 h was higher than that of Cr–31Nb. Therefore, the combination of the Cr(Nb) and Cr_2Nb phases in a eutectic mixture resulted in worse oxidation resistance than either single-phase Cr(Nb) or Cr_2Nb.

The scale formed on Cr–18.5Nb was similar to that formed on Cr–31Nb (Figure 6 vs Figure 5a). It consisted of a continuous layer of Cr_2O_3 overlying a region of mixed Cr + Nb oxides. Extensive nitridation was observed underneath the scale (Figure 6). The nitridation mimicked the underlying alloy microstructure, however the depth of nitridation was greater in the Laves phase (on the order of 100 μm) than in the Cr solid solution phase (on the order of 40–50 μm). This form of attack, in which oxidation (or nitridation, etc.) follows the underlying alloy microstructure, has been termed *in-situ* internal oxidation [15,16] because the underlying alloy microstructure is consumed in-place. Consistent with an *in-situ* process, the composition of the nitrided Cr(Nb) phase was ≈(60–65)Cr–0.5Nb–(35–40)N and the composition of the nitrided Cr_2Nb phase was ≈(40–45)Cr–(20–25)Nb–(30–35)N

After a short period of rapid weight gain, the Cr(Ta)–Cr_2Ta eutectic alloy, Cr–9.5Ta, exhibited parabolic-like oxidation kinetics similar to that observed for Cr–1Ta. However, unlike Cr–1Ta, the scale formed on Cr–9.5Ta was adherent. Therefore, the combination of the Cr(Ta) and Cr_2Ta phases in a eutectic mixture resulted in better oxidation resistance than either single-phase Cr(Ta) or Cr_2Ta.

A continuous Cr_2O_3 based scale approximately 15 μm thick was formed on Cr–9.5Ta after 24 h at 1,100°C in humid air. Extensive nitridation comparable to that observed in Cr-18.5Nb did not occur. To further investigate nitride subscale formation in Cr–9.5Ta, a coupon was exposed in an alumina crucible at 1,100°C in humid air for 120 h (six cycles) (weight change data shown in Figure 1). A continuous, adherent Cr_2O_3 scale approximately 20 μm thick formed (Figure 7). Underlying the Cr_2O_3 scale was a second layer of similar thickness which consisted of a matrix oxide phase of approximate composition 20Cr–20Ta–60O (derived from the Laves phase) and a nitride phase of approximate composition 70Cr–30N (derived from the Cr solid solution phase). Underneath this zone the Laves phase was selectively nitrided well ahead of the alloy/scale interface (on the order of 100 μm deep) whereas the Cr solid solution phase was not affected. The composition of the nitrided Cr_2Ta Laves phase was approximately 50Cr–25Ta–25N. The selective nitridation of the Laves phase mimicked the underlying alloy microstructure similar to that observed in the Cr–18.5Nb alloy. For comparative purposes with Cr–9.5Ta on a concentration basis, as-cast Cr-10Nb exposed in the same six cycle, 120 h 1,100°C screening test (weight change data shown in Figure 1) exhibited a region of nitrided Cr(Nb) and Cr_2Nb approximately 100 μm deep, underneath which lay a zone of selectively nitrided Laves phase and Cr(Nb) metal at least 100 μm deep. The depth of nitridation of the Cr(X) phase was therefore approximately five times greater in Cr–10Nb than Cr–9.5Ta. Further, the rate of metal con-

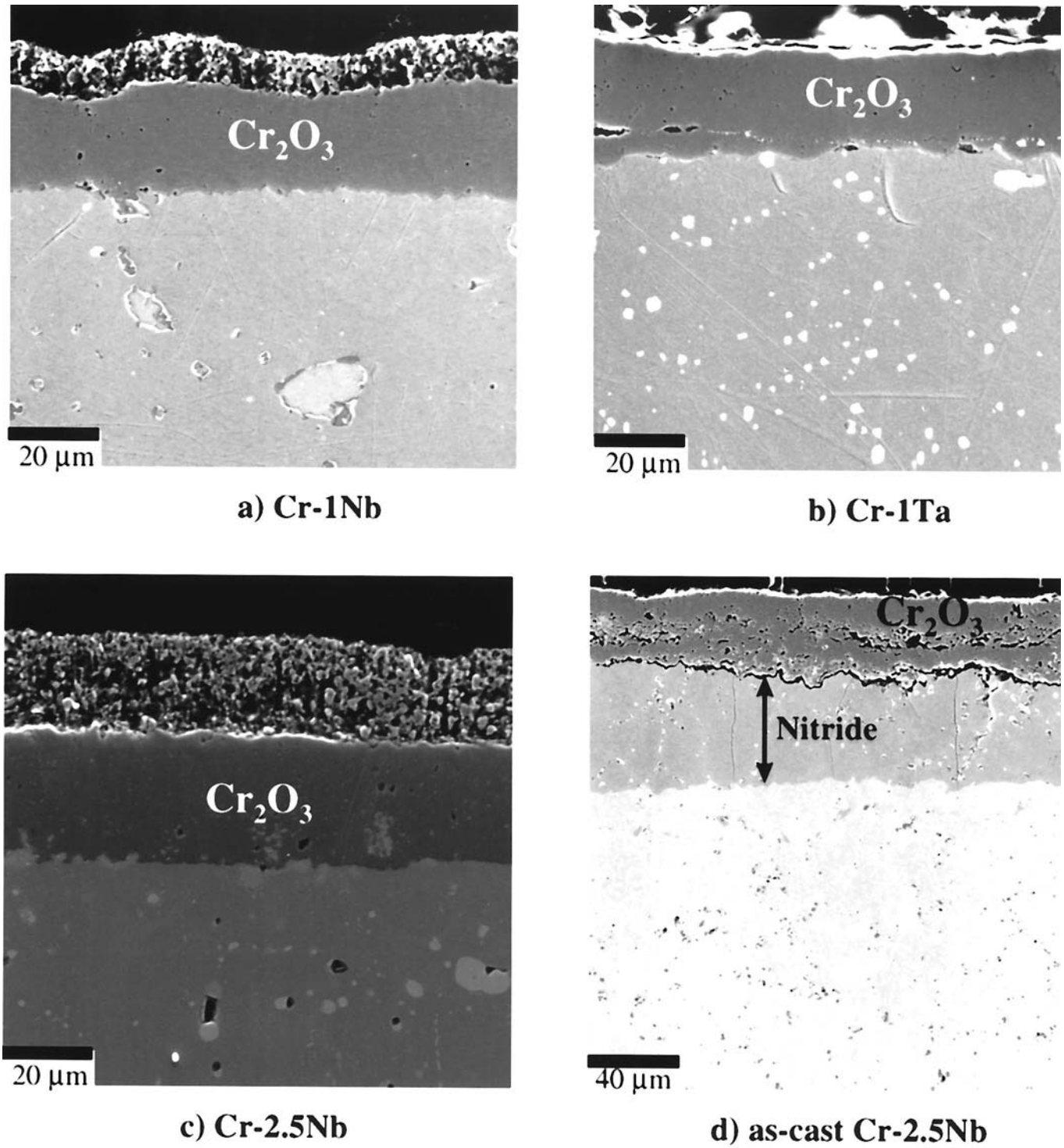


Figure 4 SEM (secondary mode) cross-section micrographs after 24 h at 1,100°C in humid air. (a) Cr–1Nb; (b) Cr–1Ta; (c) Cr–2.5Nb (TGA sample from Fig. 3); (d) as-cast Cr–2.5Nb. Note that extensive spallation occurred and that only isolated patches of scale remnants remained. The light phase visible in the alloy region of Cr-2.5Nb and Cr-1Ta is the Laves phase and/or Laves-phase based inclusions.

sumption was so severe in Cr–10Nb that the oxidation affected zones from opposite faces of the 1mm thick test coupon overlapped during the course of the test.

DISCUSSION

The superior oxidation resistance of Cr(Ta)–Cr_2Ta alloys relative to Cr(Nb)–Cr_2Nb alloys at 1,100°C in humid air can be related to two key factors: the lower amount of Laves phase in the Cr(Ta)–Cr_2Ta eutectic and the greater nitridation resistance of the Cr(Ta) phase. Both the Cr_2Nb and Cr_2Ta Laves phases exhibited poor oxidation resistance and extensive nitridation at 1,100°C in humid air. The Cr(Nb)–Cr_2Nb eutectic consisted of a Cr_2Nb Laves phase matrix and exhibited similar, poor oxidation behavior. However, the Cr(Ta)–Cr_2Ta eutectic contained only a minor amount of the Cr_2Ta phase; rather, it consisted of a Cr(Ta) matrix and exhibited relatively protective oxidation kinetics similar to that of the single-phase Cr(Ta) alloy, Cr–1Ta.

The Cr–1Ta alloy exhibited lower oxidation kinetics and, apparently, much less subscale nitridation than

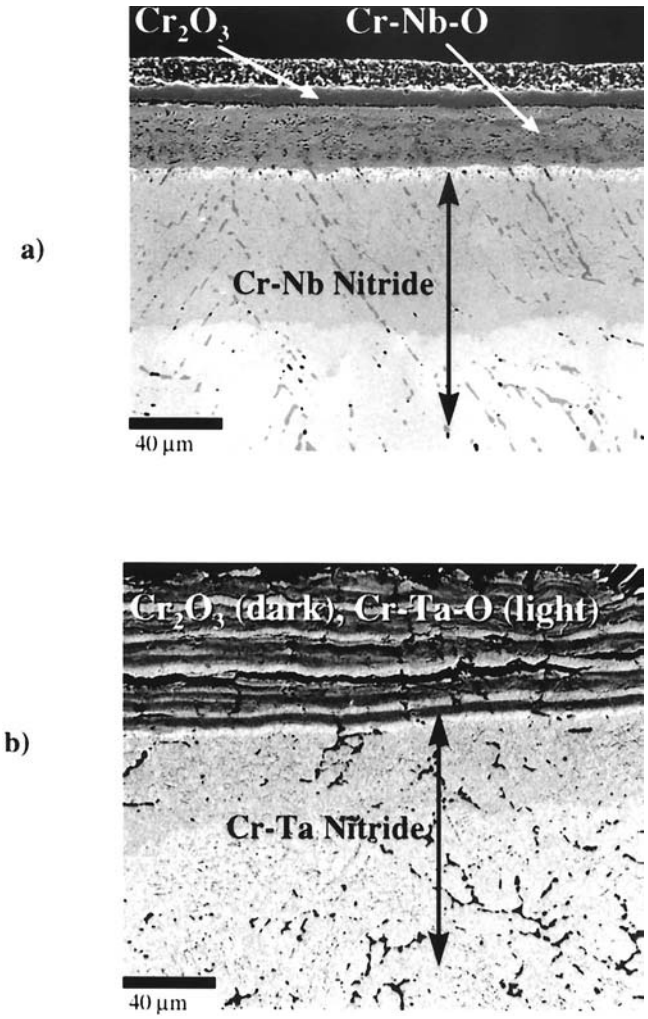


Figure 5 SEM (secondary mode) cross-section micrographs after 24 h at 1,100°C in humid air. (a) Cr–31Nb (b) Cr–31Ta. The nitrided subscale regions consisted of two layer of which only the first layer is evident in the contrast shown.

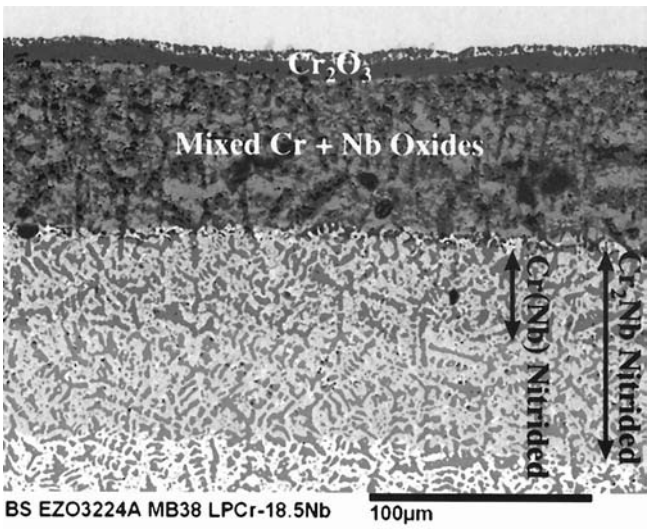


Figure 6 SEM (backscatter mode) cross-section micrograph of Cr–18.5Nb after 24 h at 1,100°C in humid air. (Laves phase light, Cr solid solution phase dark).

Cr–2.5Nb. Because some scatter in nitridation depth was observed for Cr–2.5Nb under the conditions studied, a definitive statement is not yet possible. However, this interpretation is consistent with the greater depth of nitridation observed in the Cr(Nb) phase in two-phase Cr(Nb) + Cr_2Nb alloys than in the Cr(Ta) phase in two-phase Cr(Ta)–Cr_2Ta alloys.

Why the Cr(Nb) phase should be more extensively nitrided than the Cr(Ta) phase is not clear. Because Cr–1Nb and Cr–1Ta exhibited similar behavior, any differences in nitridation depth between the Cr(Ta) and Cr(Nb) phases in the two-phase Cr(X) + Cr_2X alloys are likely related to the higher solubility of Nb in Cr than that of Ta. This may be due to the formation of a more nitrogen permeable scale, primarily as a Nb concentration effect, as there was some evidence of increased porosity in the scale formed on Cr–2.5Nb associated with local pockets of Nb-rich oxide. However, it could also result from increased nitrogen permeability in the Cr(Nb) phase due to the higher level of Nb. Longer term oxidation tests and exposure of model Cr solid solution alloys in nitrogen are planned to resolve this issue

Both detrimental and beneficial synergistic effects on the oxidation resistance of the two-phase alloys, relative to the single-phase component alloys, were observed. Consequently, oxidation resistance of two-phase alloys can't necessarily be predicted by a rule of mixture approach from their single-phase component alloys. Of particular interest are beneficial effects such as were observed for the Cr(Ta)–Cr_2Ta eutectic alloy Cr–9.5Ta.

The two-phase alloy Cr–9.5Ta exhibited oxidation kinetics comparable to the single-phase Cr(Ta) alloy, Cr–1Ta. This was not surprising since the matrix phase in Cr–9.5Ta was the Cr(Ta) phase. However, what was somewhat unexpected was the good scale adherence exhibited by Cr–9.5Ta, given that Cr–1Ta suffered from extensive scale spallation. Despite its high rate of oxidation, the single-phase Cr_2Ta alloy, Cr–31Ta, exhibited a moderate degree of scale adherence. Therefore, the two-phase alloy Cr–9.5Ta appears to have inherited the low oxidation rate of the Cr(Ta) phase and the improved scale adherence of the Cr_2Ta phase. The mechanism by which Cr_2Ta improves scale adherence is not clear, although it is known the Cr_2XLaves phase (X = Nb, Ta) has a lower thermal expansion than Cr (0.0075 cm/cm vs 0.01 cm/cm at 1,000°C [3,17]), which likely contributes to this effect. Further study of this phenomenon is planned. Finally, it should be noted that alloy development based on the Cr(Ta)–Cr_2Ta eutectic has yielded compositions with oxidation kinetics comparable to chromia-forming alloys such as INCO MA754 and Haynes 230 in testing for 1000 h (ten, 100 h cycles) at 1,100°C in air [8].

CONCLUSIONS

(1) The oxidation behavior of the two-phase Cr(X)–Cr_2X alloys can be phenomenologically understood from an examination of the oxidation behavior of single-phase alloys representing the component phases. Two-phase alloy oxidation resistance was synergistically better (Cr–Ta) or worse (Cr–Nb) than that exhibited by their single-phase component alloys, and, therefore, was not limited to a rule of mixture bracketed by the oxidation behavior of the single-phase component alloys

(2) It is postulated that two factors lead to the superior

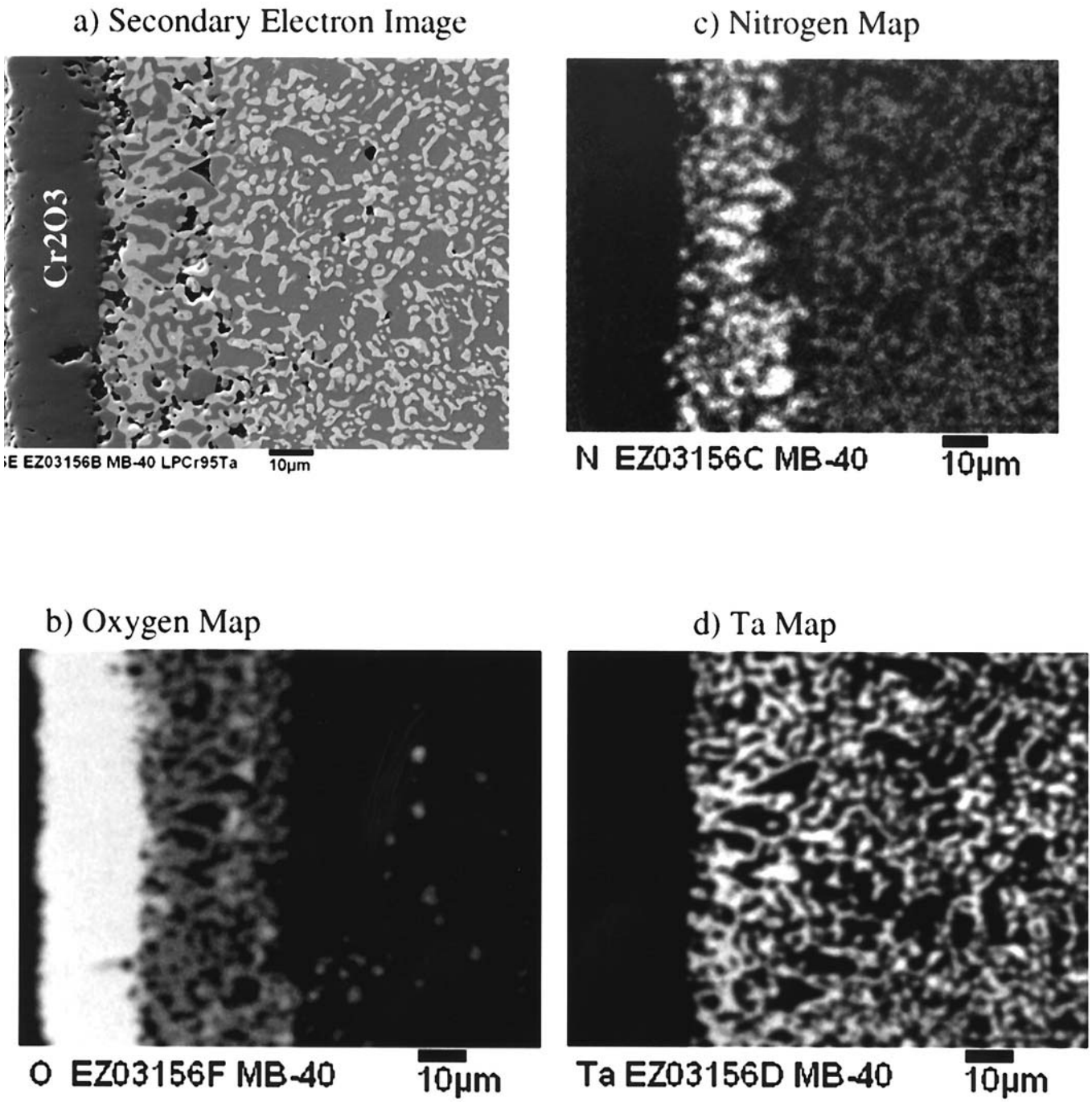


Figure 7 EPMA analysis of Cr–9.5Ta after the 6 cycle, 120 cumulative h, 1,100°C humid air exposure from Fig. 1. (a) secondary electron image (Laves phase light, Cr solid solution phase dark); (b) oxygen map; (c) nitrogen map; (d) Ta map.

oxidation resistance of the Cr(Ta)–Cr_2Ta alloys in humid air at 1,100°C relative to that of Cr(Nb)–Cr_2Nb alloys:

(a) The Cr(Ta)–Cr_2Ta eutectic contains a much lower percentage of the Laves phase than the Cr(Nb)–Cr_2Nb eutectic; the Laves phase oxidizes at a much higher rate than the Cr solid solution phase under these conditions, due in part to extensive nitridation
(b) The apparently lower extent of subscale nitridation in the Cr(Ta) phase than the Cr(Nb) phase, which was postulated to be related to the lower solubility of Ta in Cr than Nb.

ACKNOWLEDGEMENTS

The authors thank J.R. Distefano, B.A. Pint, and D.F. Wilson for comments on the manuscript. This research was sponsored by the Fossil Energy Advanced Research and Technology Development (AR&TD) Materials Program, U. S. Department of Energy, under contract DE-AC05-96OR22464 with Lockheed Martin Energy Research Corporation. The submitted manuscript has been authored by a contractor of the U.S. Government under contract No. DE-AC05-96OR22464. Accordingly, the U.S. Government retains a nonexclusive, royalty-free license to publish or reproduce the published form of this contribution, or allows others to do so for U.S. Government purposes.

REFERENCES

[1] Takeyama, M., Liu, C.T. Microstructure and mechanical-properties of Laves-phase alloys based on Cr_2Nb. *Mat. Sci. Eng.* **A132**, 61–66 (1991).

[2] Liu, C.T., Tortorelli, P.F., Horton, J.A., Carmichael, C.A. Effects of alloy additions on the microstructure and properties of Cr–Cr_2Nb alloys. *Mat. Sci. Eng. A* **214**, 23–32 (1996).

[3] Kumar, K.S., Liu, C.T. Precipitation in a Cr–Cr_2Nb alloy. *Acta Mater* **45**(9), 3671–3686 (1997).

[4] Aoyama, N., Hanada, S. Microstructure and strength of $NbCr_2$/Cr in-situ composites., *Mat. Trans. JIM* **38**(2), 155–162, (1997).

[5] Yoshida, M., Takasugi, T., High-temperature deformation of the $NbCr_2$-based Laves intermetallics in Nb–Cr–V and Nb–Cr–Mo alloy systems. *Mat. Sci. Eng. A* **224**(1–2), 77–86 (1997).

[6] Mazdiyasni, M., Miracle, D.B. In: *Intermetallic Matrix Composites*, Anton, D.L., Martin, P.L., Miracle, D.B., McMeeking, R. (Eds.), Pittsburgh, Pa: MRS, 155–162 (1990).

[7] Ravichandran, K.S., Miracle, D.B., Mendiratta, M.G. Microstructure and mechanical behavior of Cr–Cr_2Hf in situ intermetallic composites. *Metall. Mater. Trans. A* **27A**, 2583–2592 (1996).

[8] Brady, M.P., Zhu, J.H., Liu, C.T., Tortorelli, P.F., Walker, L.R., McKamey, C.G., Wright, J.L., Carmichael, C.A., Larson, D.J., Miller, M.K., Porter, W.D. Intermetallic reinforced Cr alloys for high-temperature use. *Materials at High Temperatures* **16**(4) 189–193 (1999)

[9] Brady, M.P., Zhu, J.H., Liu, C.T., Tortorelli, P.F., Walker, L.R. Oxidation resistance and mechanical properties of Laves phase reinforced Cr in-situ composites. submitted to the "Intermetallics for the Third Millennium" Special Issue of *Intermetallics* (2000, in press).

[10] Brady, M.P. and Tortorelli, P.F. In: *High Temperature Corrosion and Materials Chemistry*, Hou, P.Y., McNallan, M.J., Oltra, R., Opila, E.J., and Shores, D.A. (Eds.), ECS, Pennington, NJ., 466–476 (1998).

[11] Brady, M.P., Tortorelli, P.F., Walker, L.R. to be submitted to *Ox. Met.*

[12] Thoma, D.J. and Perepezko, J.H. An experimental evaluation of the phase-relationships and solubilities in the Nb-Cr system. *Mat. Sci. Eng.* **A156** (1992) p. 97–108.

[13] Zhu, J.H. and Liu, C.T. Oak Ridge National Laboratory, unpublished results 1997.

[14] Venkatraman, M. and Neumann, J.P. In: *Binary Alloy Phase Diagrams* Vol. 1, T.B. Massalaski, J.L. Murray, L.H. Bennett, and H. Baker eds., ASM (1986) p. 867.

[15] Gesmundo, F., Niu, Y., Viani, F., Possible scaling modes in high-temperature oxidation of 2-phase binary-alloys .2. low oxidant pressures. *Ox. Met.*, **43**(3/4), p. 379–394 (1995).

[16] Gesmundo, F. and Gleeson, B., Oxidation of multicomponent 2-phase alloys. *Ox. Met.*, **44**(1/2), p. 211 (1995).

[17] Liu, C.T. and Porter, W.D. unpublished research (1999).